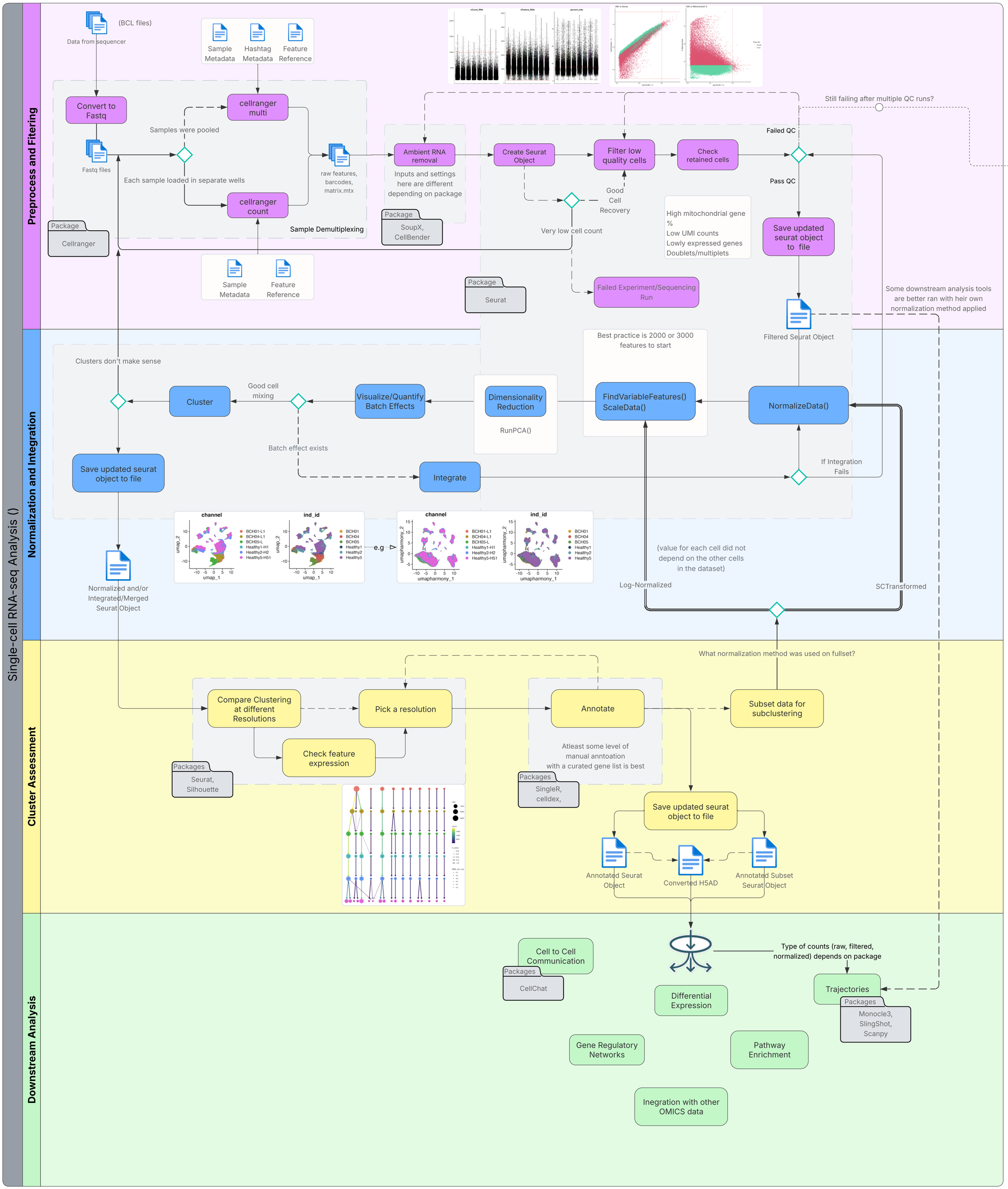
Single-cell RNA-seq Analysis ()
Preprocess and Filtering
Normalization and Integration
Cluster Assessment
Downstream Analysis
Activity diagram shape key
Start
Tool utilized
Somethings wrong threshold
Checkpoint
End
Preprocess and Filtering
Normalization and Integration
State Name
(BCL files)
Data from sequencer
Sample Metadata
Hashtag Metadata
Feature Reference
Convert to Fastq
Fastq files
Samples were pooled
Each sample loaded in separate wells
cellranger multi
cellranger count
raw features, barcodes, matrix.mtx
Sample Demultiplexing
Package
Cellranger
Sample Metadata
Feature Reference
Ambient RNA removal
Inputs and settings here are different depending on package
Package
SoupX, CellBender
Create Seurat Object
Filter low quality cells
Check retained cells
Failed QC
Pass QC
Still failing after multiple QC runs?
Review Experimental Design
Good Cell Recovery
Very low cell count
High mitochondrial gene %
Low UMI counts
Lowly expressed genes
Doublets/multiplets
Save updated seurat object to file
Failed Experiment/Sequencing Run
Package
Seurat
Filtered Seurat Object
Some downstream analysis tools are better ran with heir own normalization method applied
Best practice is 2000 or 3000 features to start
Clusters don't make sense
Cluster
Good cell mixing
Visualize/Quantify Batch Effects
Dimensionality Reduction
RunPCA()
FindVariableFeatures() ScaleData()
NormalizeData()
Batch effect exists
Save updated seurat object to file
Integrate
If Integration Fails
e.g
Normalized and/or Integrated/Merged Seurat Object
(value for each cell did not depend on the other cells in the dataset)
Log-Normalized
SCTransformed
What normalization method was used on fullset?
Compare Clustering at different Resolutions
Pick a resolution
Check feature expression
Packages
Seurat, Silhouette
Annotate
Atleast some level of manual anntoation with a curated gene list is best
Packages
SingleR, celldex,
Subset data for subclustering
Save updated seurat object to file
Annotated Seurat Object
Converted H5AD
Annotated Subset Seurat Object
Type of counts (raw, filtered, normalized) depends on package
Cell to Cell Communication
Packages
CellChat
Differential Expression
Trajectories
Packages
Monocle3, SlingShot, Scanpy
Gene Regulatory Networks
Pathway Enrichment
Inegration with other OMICS data